1. Declaring a class, data member, and function member

Q1: What keyword is used to define a class in C++?
A) struct
B) class
C) define
D) object

Q2: Which is the correct way to declare a class with a data member `int age`?
A) `int class Student { age; }` x
B) `class Student { int age; };`
C) `struct Student { age int; };` x
D) `class Student: int age {};`

Q3: In a class definition, which access specifier makes a member function or variable accessible only within the class?
A) private
B) public
C) protected
D) external

Q4: Which of the following is **not** a valid function declaration inside a class?
A) `void display();`
B) `int getData();`
C) `double calc(double x);`
D) `function displayData();`

---

2. Creating constructor and destructor

Q5: What is the main purpose of a constructor in a class?
A) To free up memory when an object goes out of scope
B) To initialize the data members of the class
C) To return the value of a function
D) To overload operators

Q6: Which of the following is **true** about constructors in C++?
A) Constructors must have a return type. x
B) Constructors can have the same name as other member functions.
C) Constructors cannot have parameters.
D) Constructors do not have a return type.

Q7: What is the correct syntax for declaring a destructor in a class?
A) `~ClassName()`
B) `ClassName()`
C) `!ClassName()`
D) `delete ClassName()`

Q8: Which of the following statements is **true** about destructors?
A) A destructor can be overloaded.
B) A destructor is automatically called when an object is destroyed.

C) Destructors cannot be virtual.
D) Destructors must be explicitly called.

---

hantar salinan nilai asal ke dalam function.

3. Pass object as function parameter

Q9: When an object is passed as a parameter to a function by **value**, what happens?
A) A reference to the object is passed.
B) The original object is modified.
C) A copy of the object is created.
D) The object is deleted after the function call.

kalau nilai berubah dlm function, salinan je ubah nilai asal tidak

Q10: Which of the following is a correct way to pass an object to a function by reference?
A) `void display(Object obj)`
B) `void display(Object &obj)`
C) `void display(Object *obj)`
D) `void display(Object &&obj)`

hantar reference object ke dalam function, bukan salinan object

kalau object berubah dlm function perubahan itu akan jadi pd object asal luar function

Q11: What is the main advantage of passing an object by reference rather than by value?
A) It guarantees the object will be modified.
B) It avoids creating a copy of the object, thus saving memory and time.
C) It ensures the object is immutable.
D) It simplifies the syntax of the function call.

---

4. Return object from a function

Q12: What happens when an object is returned from a function by **value**?
A) A reference to the object is returned.
B) A pointer to the object is returned.
C) A new copy of the object is created and returned.
D) The original object is modified directly.

Q13: Which of the following is the correct way to return an object by **reference**?
A) `Object getObject()`
B) `Object* getObject()`
C) `Object &getObject()`
D) `Object getObject&()`

Q14: If an object is returned by reference, which of the following is **true**?
A) The caller receives a copy of the object.
B) The caller can modify the original object.
C) The function cannot return a local object.
D) The object is deleted immediately after returning.

---

5. Array of class

Q15: How is an array of objects from a class declared in C++?
A) `ClassName obj[5];`

B) `ClassName obj[];`
C) `ClassName obj{5};`
D) `ClassName[5] obj;`

Q16: Which of the following is **true** about initializing an array of class objects?
A) You must initialize each object manually.
B) The constructor is called for each object in the array.
C) Only the first object is initialized, and the rest are empty.
D) Arrays of class objects cannot be initialized.

---

6. Pointer to class

Q17: What is the correct way to declare a pointer to a class object in C++?
A) `ClassName obj = &pointer;`
B) `ClassName *pointer;`
C) `pointer = new ClassName();`
D) `ClassName pointer = new obj();`

Q18: Which of the following is **true** when using a pointer to a class?
A) You use the dot (`.`) operator to access members.
B) You must deallocate memory manually using `delete` if allocated dynamically.
C) Pointers to classes cannot be used with arrays.
D) Pointers always point to a local object in the function.

Q19: What does the following line of code do?
`ClassName *ptr = new ClassName();`
A) Creates an object of type `ClassName` and stores its address in `ptr`.
B) Creates an object and returns it by reference.
C) Deletes an object of type `ClassName`.
D) Declares a constant object.

Q20: Which operator is used to access a class's member functions when using a pointer to the object?
A) `.`
B) `&`
C) `->`
D) `*`